

A Comprehensive Data-Driven Framework for Seller Lifetime Value Modeling and Behavioral Segmentation Using Probabilistic Models and Unsupervised Learning

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Abstract—E-commerce platforms increasingly depend on the sustained engagement of third-party sellers as a primary revenue source; however, existing Customer Lifetime Value (CLV) literature predominantly focuses on end consumers rather than the sellers themselves. This gap is particularly pronounced in B2B marketplace contexts, such as platforms like Olist, where independent merchants transact with the platform as a business partner, in contrast to the B2C setting that dominates prior CLV research. This study addresses this gap by adapting and applying the BG/NBD and Gamma-Gamma probabilistic models, conventionally employed for buyer-side transaction modeling, to the seller population of a marketplace platform. Using the Brazilian E-Commerce Public Dataset by Olist, we construct a Seller Lifetime Value (SeLTV) framework that estimates individual seller activity probabilities and expected transaction volumes via the BG/NBD model, and approximates per-order monetary value through the Gamma-Gamma model, from which platform commission revenue is derived as a proxy for SeLTV. These probabilistic outputs are subsequently used as features for K-Means clustering, enabling the segmentation of sellers into behaviorally distinct cohorts with tailored retention strategies. An A/B testing protocol is further proposed to validate the causal impact of cohort-specific interventions. This work demonstrates the feasibility and interpretability of repurposing established buyer-side probabilistic models for seller-side lifetime value estimation in a B2B marketplace context, contributing a scalable analytical framework for seller relationship management on digital marketplace platforms.

Index Terms—Seller Lifetime Value, BG/NBD, Gamma-Gamma, K-means, A/B Testing, Probabilistic Models, E-Commerce Analytics, Data-Driven Marketing

I. INTRODUCTION

In multi-sided e-commerce marketplaces, sellers play a critical role in sustaining platform growth because they directly determine product availability, transaction volume, and commission-based revenue. Unlike buyers, sellers are supply-side partners whose future activity affects not only short-term sales but also the long-term liquidity and competitiveness of the marketplace. Therefore, identifying which sellers are likely to remain active, generate future orders, and contribute meaningful commission revenue is an important data-driven marketing problem.

However, seller activity in an e-commerce marketplace is typically observed in a non-contractual setting. Sellers do not explicitly announce churn or cancellation; instead, inactivity is

inferred from transaction behavior. This makes traditional rule-based approaches, such as classifying sellers as inactive after a fixed number of days without orders, potentially inaccurate. A seller may simply experience a temporary sales gap, while another may have effectively stopped selling. To address this ambiguity, probabilistic customer-base models provide a more principled way to estimate future activity and lifetime value under uncertainty.

This report develops a data-driven framework for Seller Lifetime Value (SeLTV) modeling and behavioral segmentation using probabilistic models and unsupervised learning. The project adapts well-established Customer Lifetime Value (CLV) methods to the seller side of an e-commerce marketplace. Specifically, the BG/NBD model is used to estimate future seller activity, while the Gamma-Gamma model is used to estimate the expected monetary value of future seller transactions. These two components are combined with a platform commission assumption to calculate expected future seller commission revenue. Finally, K-Means clustering is applied to segment sellers into actionable groups based on predicted orders, seller activity probability, and expected commission value.

The main objective of this report is not only to forecast seller value, but also to transform predictive outputs into practical marketing and seller management strategies. Instead of treating all sellers equally, the proposed framework enables the platform to identify high-value active sellers for retention and growth, medium-potential sellers for activation and development, and low-value or low-active sellers for automated reactivation and diagnostic campaigns. In this way, the project connects predictive analytics with resource allocation, seller relationship management, and future experimental validation through A/B testing.

II. LITERATURE REVIEW

Customer Lifetime Value (CLV) is a central concept in data-driven marketing because it helps firms allocate resources based on expected future contribution rather than only historical revenue. In non-contractual settings such as retail and e-commerce, customers or sellers do not explicitly announce when they become inactive. Therefore, future activity must be inferred from transaction behavior. Schmittlein, Morrison, and

Colombo [2] introduced the Pareto/NBD framework to model customer activity and future transactions under this type of uncertainty. Later, Fader, Hardie, and Lee [1] proposed the BG/NBD model as a simpler and more practical alternative for estimating expected future transactions and the probability that an individual remains active.

While BG/NBD focuses on transaction frequency, lifetime value estimation also requires a monetary component. Fader and Hardie [3] introduced the Gamma-Gamma model to estimate the expected average monetary value of future transactions. Fader, Hardie, and Lee [4] further connected recency, frequency, and monetary value with CLV estimation, showing that behavioral transaction history can be used to compare the future value of different individuals. The combination of BG/NBD and Gamma-Gamma has also been applied in practical CLV prediction contexts, such as financial customer value estimation [5].

Although these models were originally developed for customer-base analysis, this report adapts the same probabilistic logic to the seller side of an e-commerce marketplace. In this setting, a seller's future value depends on expected future orders, expected monetary value per order, and the platform's commission rate. Therefore, the report defines Seller Lifetime Value (SeLTV) as expected future commission revenue generated by each seller.

After estimating seller activity and value, segmentation is needed to translate predictive outputs into actionable groups. Because seller-level monetary and behavioral variables are often skewed, this report applies the Yeo-Johnson transformation proposed by Yeo and Johnson [6] before clustering. K-Means clustering is then used to group sellers based on predicted orders, seller activity probability, and expected commission revenue. To evaluate the quality of the clustering result, the Silhouette method proposed by Rousseeuw [7] is used together with the Elbow Method. The resulting segments support differentiated seller management strategies, including retention, activation, and low-cost reactivation.

III. EXPLORATORY DATA ANALYSIS

a) Data Description: This study uses the Brazilian E-Commerce Public Dataset by Olist, together with the marketing funnel datasets, to analyze seller behavior on a marketplace platform. The core transactional data includes orders, order items, sellers, customers, products, payments, reviews, and geolocation information. The marketing funnel data provides additional information on marketing-qualified leads and closed seller deals, which is used only as descriptive context for seller acquisition.

The main analytical unit of this study is the seller, rather than the end customer. Therefore, the raw item-level and order-level data are transformed into seller-order transactions. Each seller-order transaction represents a completed commercial activity by a seller on the platform and is later used to construct frequency, recency, tenure, monetary value, and future activity variables.

TABLE I
DATASET DESCRIPTION AND ROLE IN ANALYSIS

Dataset	Role in Analysis
orders	Order status and purchase timestamps.
order_items	Seller ID, product ID, price, freight value, and seller-order records.
sellers	Seller geographic information.
customers	Customer geographic information for demand context.
products	Product category information.
payments	Payment value and payment method context.
reviews	Service quality and customer satisfaction signals.
geolocation	Geographic distribution analysis.
mql.csv	Lead information, including contact date, landing page, and acquisition origin.
closed.csv	Converted seller lead information, attributes.

b) Data Overview and Marketplace Dynamics: The Olist marketplace contains a large and diverse transaction base, with approximately 98.8K orders, 95.6K unique customers, 3.1K active sellers, and total revenue of around 15.42M during the 2017–2018 period. The average order value is approximately 137.43, indicating that the dataset is sufficiently rich for seller-level value modeling.

The order volume and revenue trend shows clear marketplace growth from 2017 to 2018. Orders and revenue move in the same direction, suggesting that platform growth is mainly driven by increasing transaction activity rather than only by higher order values. This temporal growth pattern supports the use of a chronological calibration-holdout split instead of a random split, because future market conditions should not leak into model training.

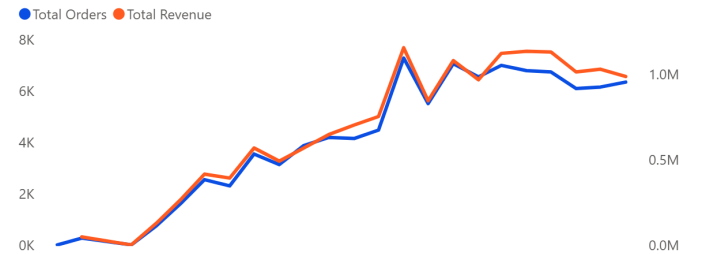


Fig. 1. Order Volume and Revenue Trend, 2017–2018

c) Seller Geography and Acquisition Context: The seller base is geographically broad but highly concentrated. The seller dashboard reports 3.095K total sellers, covering 23 states and 611 cities. São Paulo is the dominant selling state, while São Paulo city is also the largest selling city. The dashboard's seller analysis view shows seller distribution by state, seller distribution by city, revenue concentration by seller, and total revenue by state.

This geographic concentration suggests that Olist's seller supply is strongly anchored in the Southeast and South of Brazil. Such concentration may reflect stronger logistics infrastructure, denser commercial networks, and higher seller acquisition efficiency in these regions. However, it also implies that platform performance may be sensitive to regional competition and operational bottlenecks.

d) Seller Performance Heterogeneity: The most important EDA finding is the strong heterogeneity in seller performance. Seller-level order count, total GMV, active months, and seller revenue contribution are highly right-skewed. Most sellers generate only a small number of orders, while a smaller

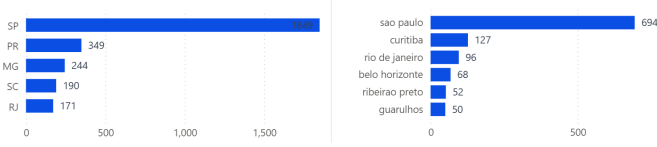


Fig. 2. Seller Distribution by State and City

group of high-frequency sellers contributes disproportionately to platform activity and revenue.

This long-tail structure is central to the modeling design. Since sellers differ substantially in activity level and monetary contribution, simple averages are not sufficient to estimate future seller value. The observed heterogeneity supports the use of BG/NBD to model future transaction frequency and Gamma-Gamma to estimate expected monetary value.

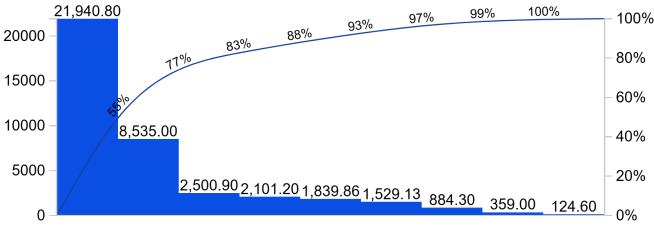


Fig. 3. Revenue Concentration by Seller

IV. METHODOLOGY

A. Data Preprocessing and Feature Construction

1) *Data Cleaning and Transaction Aggregation*: This study utilizes the *Brazilian E-Commerce Public Dataset by Olist* to construct a seller-centric transactional database for Seller Lifetime Value (SeLTV) modeling. Since the objective of the study is to evaluate seller behavior from the perspective of a marketplace platform, the raw order-level and item-level datasets were integrated to create a unified seller-order transaction table.

The preprocessing stage began by merging the `orders`, `order_items`, and `sellers` tables using the shared identifiers `order_id` and `seller_id`. To ensure transactional consistency and economic validity, several filtering criteria were applied. Only transactions associated with delivered orders were retained, as these represent successfully completed commercial activities capable of generating realized platform revenue. Observations with missing seller identifiers, order identifiers, transaction timestamps, or product prices were removed. In addition, transactions with non-positive monetary values were excluded to avoid distortions in subsequent monetary estimation procedures.

Formally, a transaction record is retained only if it satisfies the following validity constraints:

```

order_status = "delivered"
price > 0
seller_id ≠ ∅
order_id ≠ ∅
transaction_date ≠ ∅

```

The transaction timestamp was defined using the order purchase datetime (`order_purchase_timestamp`), which serves as the temporal reference for all behavioral analyses. Gross Merchandise Value (GMV) was approximated using the product-level price field provided in the dataset.

To align the data structure with probabilistic customer-base modeling requirements, the cleaned item-level records were subsequently aggregated into seller-order transactions. Specifically, all items associated with the same seller and order were consolidated into a single transaction event. For each seller-order pair, the following aggregated variables were computed:

- Transaction date (minimum purchase timestamp)
- Total GMV
- Total freight value
- Number of items
- Seller geographic information (city and state)

The aggregation process transforms granular item-level observations into seller-level transactional sequences appropriate for Buy-Till-You-Die (BTYD) modeling frameworks such as BG/NBD and Gamma-Gamma. This transformation is particularly important because the BG/NBD model assumes discrete transaction occurrences over time at the entity level, which in this study corresponds to sellers operating on the marketplace platform.

2) *Calibration and Holdout Split*: To evaluate the predictive performance of the proposed probabilistic framework, the transaction data were partitioned into calibration and holdout periods following the conventional BTYD modeling paradigm.

The observation window begins on January 1, 2017 and includes all eligible transactions up to the maximum holdout horizon. The calibration period ends on April 30, 2018 at 23:59:59, while the holdout period extends beyond the calibration boundary for multiple forecasting horizons.

Let t_c = April 30, 2018 denote the calibration cutoff date. Transactions satisfying:

$$\text{transaction_date} \leq t_c$$

were assigned to the calibration dataset, whereas transactions occurring after the cutoff date were assigned to the holdout dataset:

$$t_c < \text{transaction_date} \leq t_c + h$$

where $h \in \{30, 60, 90\}$ days represents the prediction horizon.

The calibration dataset is used for parameter estimation of the BG/NBD and Gamma-Gamma models, while the

holdout dataset enables out-of-sample evaluation of transaction frequency and monetary value predictions. The use of multiple forecast horizons allows the robustness of the proposed framework to be examined across short-, medium-, and longer-term seller activity windows.

This temporal splitting strategy preserves the chronological structure of marketplace transactions and prevents information leakage from future observations into the training process.

3) *Model Feature Construction*: Following the calibration split, seller-level behavioral features were constructed to support probabilistic transaction modeling. The feature engineering procedure follows the standard Buy-Till-You-Die (BTYD) framework while adapting its interpretation to the seller-side marketplace context.

For each seller i , the following variables were computed from the calibration-period transactions:

a) *Transaction Frequency*: The BG/NBD model defines transaction frequency as the number of repeat transactions completed by a seller during the calibration period. Let:

$$x_i = n_i - 1 \quad (1)$$

where n_i denotes the total number of observed seller transactions. This definition excludes the first observed transaction because the BTYD framework models repeat purchasing behavior conditional on acquisition.

b) *Recency*: Recency measures the elapsed time between the seller's first and most recent observed transactions during the calibration period:

$$\text{Recency}_i = t_{i,\text{last}} - t_{i,\text{first}} \quad (2)$$

Recency captures the temporal concentration of seller activity and provides important information regarding seller engagement intensity.

c) *Customer Age (T)*: The variable T_i represents the total observation duration between the seller's first transaction and the end of the calibration period:

$$T_i = t_c - t_{i,\text{last}} \quad (3)$$

where t_c denotes the calibration cutoff date. To ensure numerical stability during model estimation, extremely small values of T_i were constrained to a small positive constant.

d) *Monetary Value*: For Gamma-Gamma estimation, the monetary value variable was defined as the average GMV of repeat transactions only:

$$m_i = \frac{1}{x_i} \sum_{j=2}^{n_i} \text{GMV}_{ij} \quad (4)$$

This formulation is consistent with the assumptions of the Gamma-Gamma model, which estimates expected transaction value conditional on repeat purchasing behavior. Sellers without repeat transactions were assigned a monetary value of zero during preprocessing.

e) *Additional Descriptive Features*: Several supplementary variables were also retained for descriptive and downstream analytical purposes, including Total calibration-period orders, Total calibration-period GMV, Historical average order value (AOV), First transaction date, Last transaction date.

These seller-level features constitute the foundational inputs for subsequent BG/NBD activity forecasting and Gamma-Gamma monetary estimation procedures.

B. Seller Activity Modeling Using BG/NBD

The BG/NBD model, originally proposed by Fader, Hardie, and Lee [1], serves as the foundation for seller activity modeling in this framework. While the model was originally developed to predict repeat-purchase behavior of end consumers in non-contractual settings, we adapt and apply it to the seller population of the Olist marketplace, treating each seller as an independent entity whose transaction history with the platform is summarized by a Recency-Frequency-Tenure (R-F-T) triple.

1) *BG/NBD Model Assumptions*: The BG/NBD model is built upon five behavioral assumptions, restated here in the context of seller-side transactions:

a) *Poisson transaction process*: While a seller remains active, the number of fulfilled orders in a time period of length t follows a Poisson process with individual transaction rate λ . Equivalently, the inter-order time for an active seller is exponentially distributed with rate λ .

b) *Gamma heterogeneity in transaction rates*: Transaction rates λ vary across sellers according to a Gamma distribution with shape parameter r and scale parameter α :

$$f(\lambda \mid r, \alpha) = \frac{\alpha^r \lambda^{r-1} e^{-\lambda\alpha}}{\Gamma(r)}, \quad \lambda > 0 \quad (5)$$

c) *Geometric dropout process*: After each fulfilled order, a seller may permanently cease activity with probability p . The point of dropout is therefore geometrically distributed across transactions:

$$P(\text{inactive after } j\text{-th order}) = p(1-p)^{j-1}, \quad j = 1, 2, 3, \dots \quad (6)$$

d) *Beta heterogeneity in dropout probabilities*: Dropout probabilities p vary across sellers according to a Beta distribution with shape parameters a and b :

$$f(p \mid a, b) = \frac{p^{a-1}(1-p)^{b-1}}{B(a, b)}, \quad 0 \leq p \leq 1 \quad (7)$$

e) *Independence of λ and p* : The transaction rate and dropout probability vary independently across sellers.

Prior to model fitting, a set of assumption checks is conducted on the calibration dataset to verify that the Olist seller transaction data is broadly consistent with the above conditions. These checks include validating the structural integrity of the R-F-T summary, examining the coefficient of variation of inter-order times to assess the Poisson process assumption, inspecting the frequency distribution for evidence of Gamma-consistent heterogeneity, and evaluating the stationarity of seller-level order rates across the early and late subperiods of calibration.

2) *Expected Number of Orders Estimation:* Each seller is represented by the summary tuple (x, t_x, T) , where x denotes the number of repeat orders observed in the calibration period $(0, T]$ and t_x is the time of the most recent order. The four model parameters (r, α, a, b) are estimated by maximizing the sample log-likelihood function:

$$\ell(r, \alpha, a, b) = \sum_{i=1}^N \ln L(r, \alpha, a, b \mid x_i, t_{x_i}, T_i) \quad (8)$$

where the individual-level likelihood is:

$$L(r, \alpha, a, b \mid x, t_x, T) = \frac{B(a, b+x)}{B(a, b)} \frac{\Gamma(r+x) \alpha^r}{\Gamma(r)(\alpha+T)^{r+x}} + \delta_{x>0} \frac{B(a+1, b+x-1)}{B(a, b)} \frac{\Gamma(r+x) \alpha^r}{\Gamma(r)(\alpha+t_x)^{r+x}} \quad (9)$$

Optimization is performed using the `BetaGeoFitter` implementation from the `lifetimes` Python library. Once parameters are estimated, the expected number of future orders for seller i over a prediction horizon of length t is:

$$\mathbb{E}[Y(t) \mid x, t_x, T, r, \alpha, a, b] = \frac{a+b+x-1}{a-1} \left[1 - \left(\frac{\alpha+T}{\alpha+T+t} \right)^{r+x} \right] \times {}_2F_1 \left(r+x, b+x; a+b+x-1; \frac{t}{\alpha+T+t} \right) \Bigg/ \left[1 + \delta_{x>0} \frac{a}{b+x-1} \left(\frac{\alpha+T}{\alpha+t_x} \right)^{r+x} \right] \quad (10)$$

where ${}_2F_1(\cdot)$ is the Gaussian hypergeometric function and $\delta_{x>0} = 1$ if $x > 0$, 0 otherwise. Predictions are generated across three forward-looking horizons $t \in \{30, 60, 90\}$ days to assess the sensitivity of downstream SeLTV estimates to the choice of prediction window.

In addition to forecasting future order volumes, the BG/NBD model yields the probability that a seller is still active (i.e., has not yet permanently dropped out) at the end of the calibration period T . This quantity is derived analytically from the model's posterior likelihood structure as:

$$P(\text{active} \mid x, t_x, T, r, \alpha, a, b) = \frac{1}{1 + \delta_{x>0} \frac{a}{b+x-1} \left(\frac{\alpha+T}{\alpha+t_x} \right)^{r+x}} \quad (11)$$

3) *Seller Active Probability Derivation:* In addition to the long-run activity status captured by Equation (11), a complementary forward-looking engagement probability is derived to quantify the likelihood that a seller places at least one order within a future horizon of length t . Given that the expected number of future orders $\mathbb{E}[Y_i(t)]$ is obtained from the BG/NBD model under a Poisson transaction process, the probability of observing at least one transaction in the period $(T, T+t]$ can be approximated as:

$$\hat{p}_i(t) = 1 - e^{-\mathbb{E}[Y_i(t)]} \quad (12)$$

This approximation follows directly from the zero-count probability of a Poisson random variable with mean $\mathbb{E}[Y_i(t)]$. Unlike $P(\text{active} \mid x, t_x, T)$, which reflects the posterior probability that a seller has not yet permanently churned at the end of the calibration period, $\hat{p}_i(t)$ is a horizon-conditional quantity that additionally accounts for the possibility that an active seller may simply not transact within the specified window.

4) Evaluation Metrics:

a) *Holdout Target Construction:* Model evaluation is conducted under a train-holdout split scheme. The calibration period spans from January 2017 through April 2018, during which the R-F-T summary for each seller is constructed. Three holdout windows of $t \in \{30, 60, 90\}$ days are then defined immediately following the calibration end. For each seller i and horizon t , the actual number of orders placed in the holdout window is computed as:

$$y_i^{(t)} = |\{j : \text{order}_{ij} \text{ placed in } (T_{\text{cal}}, T_{\text{cal}} + t)\}| \quad (13)$$

The binary activity indicator is then derived as:

$$a_i^{(t)} = \mathbf{1}[y_i^{(t)} > 0] \quad (14)$$

which equals 1 if seller i places at least one order during the holdout window, and 0 otherwise.

b) *Order Volume Metrics:* To assess the accuracy of the BG/NBD model in predicting the number of future orders $\hat{y}_i^{(t)} = \mathbb{E}[Y_i(t)]$ at the seller level, we compute the following metrics over the full evaluation population $\mathcal{S}$:

(i) Mean Absolute Error (MAE):

$$\text{MAE}_{\text{orders}}^{(t)} = \frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} |\hat{y}_i^{(t)} - y_i^{(t)}| \quad (15)$$

(ii) Median Absolute Error (MedianAE):

$$\text{MedianAE}_{\text{orders}}^{(t)} = \text{median}_{i \in \mathcal{S}} |\hat{y}_i^{(t)} - y_i^{(t)}| \quad (16)$$

(iii) Weighted Absolute Percentage Error (WAPE):

$$\text{WAPE}_{\text{orders}}^{(t)} = \frac{\sum_{i \in \mathcal{S}} |\hat{y}_i^{(t)} - y_i^{(t)}|}{\sum_{i \in \mathcal{S}} y_i^{(t)}} \quad (17)$$

WAPE is preferred over MAPE in this setting because many sellers have $y_i^{(t)} = 0$, which renders percentage-based errors undefined at the individual level. WAPE instead measures the total absolute prediction error as a fraction of the total actual order volume across the seller population, yielding a pooled, volume-weighted error rate.

c) *Seller Activity Classification Metrics:* The expected number of future orders $\hat{y}_i^{(t)}$ is used as a continuous ranking score to evaluate the model's ability to identify which sellers will remain active during the holdout period. Rather than applying a fixed classification threshold, we treat the problem as a ranking task and evaluate using:

(i) ROC-AUC: The area under the Receiver Operating Characteristic curve measures the probability that a randomly

selected active seller is ranked higher than a randomly selected inactive seller by the model's scoring function.

$$\text{ROC-AUC}^{(t)} = P\left(\hat{y}_i^{(t)} > \hat{y}_j^{(t)} \mid a_i^{(t)} = 1, a_j^{(t)} = 0\right) \quad (18)$$

- (ii) PR-AUC: The area under the Precision-Recall curve measures ranking quality under class imbalance, placing greater emphasis on the model's ability to correctly rank truly active sellers at the top of the list. PR-AUC is particularly relevant in contexts where the active seller population is smaller than the inactive population.
- (iii) Lift at Top- k : For $k \in \{0.10, 0.20\}$, the lift at top- k measures the enrichment of active sellers among the top k fraction of sellers ranked by $\hat{y}_i^{(t)}$ relative to the population base rate:

$$\text{Lift}_k^{(t)} = \frac{\frac{1}{|\mathcal{S}_k|} \sum_{i \in \mathcal{S}_k} a_i^{(t)}}{\frac{1}{|\mathcal{S}|} \sum_{i \in \mathcal{S}} a_i^{(t)}} \quad (19)$$

where $\mathcal{S}_k$ denotes the top- k fraction of sellers ordered by descending predicted order count. A lift of L at top-10% indicates that focusing on the top-ranked 10% of sellers yields L times the active seller density compared to random selection.

C. Monetary Value Modeling Using Gamma-Gamma

1) *Gamma-Gamma Model Assumptions*: To estimate the expected monetary value of each seller's future transactions, we employ the Gamma-Gamma model of spend per transaction as presented by Fader and Hardie [3]. The model characterizes the cross-sectional distribution of average transaction values under three core assumptions:

a) *Gamma-distributed transaction values*: The monetary value z_i of each individual order placed by a seller is assumed to follow a Gamma distribution with shape parameter p and rate parameter ν , yielding a latent mean transaction value $\zeta = p/\nu$. For a seller with x repeat orders, the observed average order value $\bar{z} = \sum_{i=1}^x z_i/x$ follows a $\text{Gamma}(px, \nu x)$ distribution.

b) *Gamma heterogeneity in mean transaction values*: The latent mean transaction value ζ varies across sellers through a Gamma mixing distribution on ν , i.e., $\nu \sim \text{Gamma}(q, \gamma)$, giving rise to an inverse-Gamma marginal distribution of ζ with mean:

$$\mathbb{E}[Z \mid p, q, \gamma] = \frac{p\gamma}{q-1} \quad (20)$$

c) *Independence of monetary value and transaction process*: The distribution of average transaction values across sellers is assumed to be independent of the underlying transaction process. This assumption is empirically verified prior to model fitting by examining the Pearson and Spearman correlations between seller repeat frequency and average order value, both on raw and log-transformed scales.

Prior to model fitting, assumption checks are conducted to verify that the monetary value data satisfies the structural

requirements of the model. These checks include confirming the absence of non-positive monetary values, assessing the right-skewed shape of the AOV distribution consistent with a Gamma form, verifying the independence between frequency and monetary value, and evaluating the temporal stability of seller-level AOV across the calibration period.

2) *Expected Average Order Value Estimation*: The Gamma-Gamma model parameters (p, q, γ) are estimated by maximizing the sample log-likelihood:

$$\ell(p, q, \gamma) = \sum_{i=1}^I \ln f(\bar{z}_i \mid p, q, \gamma; x_i) \quad (21)$$

where the marginal density of $\bar{z}$ given x repeat transactions is:

$$f(\bar{z} \mid p, q, \gamma; x) = \frac{\Gamma(px+q)}{\Gamma(px)\Gamma(q)} \cdot \frac{\bar{z}^{px-1} x^{px} \gamma^q}{(\gamma + x\bar{z})^{px+q}} \quad (22)$$

Estimation is restricted to sellers with at least one repeat transaction ($x > 0$) and a positive monetary value, using the `GammaGammaFitter` implementation from the `lifetimes` library. For sellers ineligible for Gamma-Gamma estimation (i.e., one-time sellers with $x = 0$), the conditional expectation reduces to the population mean $\mathbb{E}[Z]$, and their observed historical AOV is used as a heuristic fallback.

Once the parameters are estimated, the posterior expected average order value for each seller is computed via Bayesian shrinkage toward the population mean:

$$\begin{aligned} \mathbb{E}[Z \mid \bar{z}, x, p, q, \gamma] &= \frac{p(\gamma + x\bar{z})}{px + q - 1} \\ &= \left(\frac{q-1}{px + q - 1} \right) \frac{p\gamma}{q-1} + \left(\frac{px}{px + q - 1} \right) \bar{z} \end{aligned} \quad (23)$$

This expression is a convex combination of the population mean and the seller's own observed AOV, where the weight placed on the individual observation increases monotonically with the number of transactions x . This shrinkage property is particularly well suited for sparse transaction histories, which are prevalent in the Olist seller population.

3) *Seller Lifetime Value Derivation*: Seller Lifetime Value (SeLTV) is defined as the total platform commission revenue attributable to a given seller over a future prediction horizon t . Combining the outputs of the BG/NBD and Gamma-Gamma models under the independence assumption between transaction timing and monetary value [1], SeLTV is formulated as:

$$\text{SeLTV}_i(t) = \mathbb{E}[Y_i(t)] \times \mathbb{E}[Z_i \mid \bar{z}_i, x_i] \times r_c \quad (24)$$

where $\mathbb{E}[Y_i(t)]$ is the expected number of future orders from Equation (10), $\mathbb{E}[Z_i \mid \bar{z}_i, x_i]$ is the Bayesian-adjusted expected AOV from Equation (23), and r_c is the platform's commission rate which is the share of GMV retained as revenue by the marketplace.

This formulation differs fundamentally from conventional buyer-side CLV in that the unit of value is not consumer expenditure but platform commission revenue generated by

seller activity. The commission-based proxy is justified on the grounds that marketplace revenue is structurally proportional to seller-generated GMV, obviating the need for a per-category margin model. Furthermore, by grounding SeLTV in probabilistic estimates of both future order volume and expected monetary value, the framework naturally propagates uncertainty from both model components into the final value estimate, enabling more nuanced downstream segmentation than a purely historical heuristic would permit.

4) Evaluation Metrics:

a) *Holdout Target Construction:* For each seller i and holdout horizon t , the actual average order value (AOV) is computed exclusively over sellers who placed at least one order during the holdout window, i.e., sellers with $y_i^{(t)} > 0$:

$$\bar{z}_i^{(t)} = \frac{\sum_{j=1}^{y_i^{(t)}} z_{ij}^{(t)}}{y_i^{(t)}} = \frac{G_i^{(t)}}{y_i^{(t)}} \quad (25)$$

where $G_i^{(t)}$ is the total GMV generated by seller i during the holdout window and $z_{ij}^{(t)}$ is the value of the j -th individual order. The restriction to sellers with $y_i^{(t)} > 0$ is necessary because AOV is undefined for sellers who generate no orders in the holdout period.

Similarly, the actual commission generated by seller i over horizon t is defined as:

$$c_i^{(t)} = G_i^{(t)} \times r_c \quad (26)$$

and the predicted commission is:

$$\hat{c}_i^{(t)} = \hat{y}_i^{(t)} \times \mathbb{E}[Z_i \mid \bar{z}_i, x_i] \times r_c \quad (27)$$

b) *AOV Accuracy Metrics:* The Gamma-Gamma model is evaluated on its ability to estimate the expected AOV $\mathbb{E}[Z_i \mid \bar{z}_i, x_i]$ for each seller. Evaluation is conducted on the subset of sellers with holdout orders $y_i^{(t)} > 0$, denoted $\mathcal{S}_+^{(t)} = \{i : y_i^{(t)} > 0\}$:

(i) MAE (AOV):

$$\text{MAE}_{\text{AOV}}^{(t)} = \frac{1}{|\mathcal{S}_+^{(t)}|} \sum_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right| \quad (28)$$

(ii) Median AE (AOV):

$$\text{MedianAE}_{\text{AOV}}^{(t)} = \text{median}_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right| \quad (29)$$

(iii) WAPE (AOV):

$$\text{WAPE}_{\text{AOV}}^{(t)} = \frac{\sum_{i \in \mathcal{S}_+^{(t)}} \left| \mathbb{E}[Z_i] - \bar{z}_i^{(t)} \right|}{\sum_{i \in \mathcal{S}_+^{(t)}} \bar{z}_i^{(t)}} \quad (30)$$

(iv) Spearman Rank Correlation (AOV): Since the primary use of AOV estimates is to rank sellers by expected

monetary value, we additionally compute the Spearman rank correlation between predicted and actual AOV:

$$\rho_{\text{AOV}}^{(t)} = \text{Spearman}(\mathbb{E}[Z_i], \bar{z}_i^{(t)}), \quad i \in \mathcal{S}_+^{(t)} \quad (31)$$

A high Spearman correlation indicates that the model correctly orders sellers by their monetary value, even if absolute estimates are imperfect which is a property that is sufficient for the downstream segmentation task.

c) *Commission Accuracy Metrics:* The final SeLTV estimate $\hat{c}_i^{(t)}$ integrates both model components. Its accuracy is evaluated across the full seller population $\mathcal{S}$, using the same family of metrics applied to orders:

(i) MAE and Median AE (commission): Point-error measures at the individual seller level, analogous to the order-level definitions above.

(ii) WAPE (commission):

$$\text{WAPE}_{\text{comm}}^{(t)} = \frac{\sum_{i \in \mathcal{S}} \left| \hat{c}_i^{(t)} - c_i^{(t)} \right|}{\sum_{i \in \mathcal{S}} c_i^{(t)}} \quad (32)$$

(iii) Spearman Rank Correlation (commission): The Spearman rank correlation between predicted and actual commission evaluates whether the model correctly prioritizes sellers by their revenue contribution to the platform:

$$\rho_{\text{comm}}^{(t)} = \text{Spearman}(\hat{c}_i^{(t)}, c_i^{(t)}), \quad i \in \mathcal{S} \quad (33)$$

This metric is central to evaluating the usefulness of SeLTV for marketing resource allocation, since a high rank correlation implies that directing effort toward high-SeLTV sellers reliably captures a disproportionate share of actual platform commission.

D. Seller Segmentation Using K-Means Clustering

1) Feature Selection, Scaling, and Transformation:

a) *Feature Selection:* The feature set for clustering is constructed exclusively from the probabilistic outputs of the BG/NBD and Gamma-Gamma models. The prediction horizon used as input to the segmentation stage will be selected based on the model evaluation results: specifically, the horizon that yields the best trade-off across order volume accuracy, seller activity classification quality, and commission ranking performance is adopted as the basis for all downstream clustering features. Three features are constructed at the chosen horizon t^* :

(i) $\hat{y}_i^{(t^*)}$: the expected number of orders over the selected horizon, derived from the BG/NBD conditional expectation (Equation (10)).

(ii) $\hat{p}_i^{(t^*)}$: an approximate probability that seller i places at least one order within the horizon, defined as:

$$\hat{p}_i^{(t^*)} = 1 - e^{-\hat{y}_i^{(t^*)}} \quad (34)$$

This Poisson-based approximation serves as a continuous activity score, capturing the seller's likelihood of remaining engaged with the platform.

- (iii) $\hat{c}_i^{(t^*)}$: the expected commission revenue attributable to seller i over the selected horizon, as defined in Equation (24).

These three features jointly encode a seller's expected activity intensity ($\hat{y}_i^{(t^*)}$), engagement probability ($\hat{p}_i^{(t^*)}$), and revenue potential ($\hat{c}_i^{(t^*)}$) which are the three dimensions most relevant to platform-level seller management decisions. Importantly, all three are forward-looking model outputs rather than raw historical statistics, ensuring that the resulting segments reflect predicted future behavior rather than past behavior alone.

b) Transformation and Standardization: The raw feature distributions are highly right-skewed, as a small number of high-frequency sellers with large transaction volumes dominate all three features. Applying K-Means directly to such distributions would cause centroids to be heavily pulled toward outliers, yielding clusters that reflect scale effects rather than genuine behavioral differences.

To address this, we apply the Yeo-Johnson power transformation [6] followed by standardization to zero mean and unit variance, using the `PowerTransformer` from `scikit-learn` with `standardize=True`. The Yeo-Johnson family is chosen over the Box-Cox transformation because it accommodates zero-valued observations (i.e., sellers with $\hat{y}_i^{(t^*)} \approx 0$), which are prevalent in the dataset. Formally, for a feature value v , the transformation is:

$$\psi_\lambda(v) = \begin{cases} \frac{(v+1)^\lambda - 1}{\lambda}, & \lambda \neq 0, v \geq 0 \\ \ln(v+1), & \lambda = 0, v \geq 0 \\ -\frac{(-v+1)^{2-\lambda} - 1}{2-\lambda}, & \lambda \neq 2, v < 0 \\ -\ln(-v+1), & \lambda = 2, v < 0 \end{cases} \quad (35)$$

where the optimal λ is estimated via maximum likelihood separately for each feature. The transformed features are then standardized to produce the final input matrix $\mathbf{X}_{\text{scaled}} \in \mathbb{R}^{N \times 3}$, on which K-Means is fitted.

2) Determination of Optimal Number of Clusters: Selecting the appropriate number of clusters K requires balancing intra-cluster compactness against inter-cluster separation. We employ a two-stage procedure: (i) the Elbow method to identify a candidate range of K , followed by (ii) the Silhouette criterion to select the optimal value within that range.

a) Stage 1 - Elbow Method: K-Means is fitted for $K \in \{2, 3, \dots, K_{\text{max}}\}$ and the within-cluster sum of squared distances (inertia) is recorded for each:

$$\mathcal{J}(K) = \sum_{k=1}^K \sum_{i \in \mathcal{C}_k} \|\mathbf{x}_i - \boldsymbol{\mu}_k\|^2 \quad (36)$$

where $\mathcal{C}_k$ is the set of sellers assigned to cluster k and $\boldsymbol{\mu}_k$ is the corresponding centroid. As K increases, $\mathcal{J}(K)$ decreases monotonically; however, the rate of decrease typically exhibits a pronounced "elbow" that is a point of inflection beyond which additional clusters yield diminishing marginal reduction in inertia. The elbow chart is used to identify a range of

plausible values of K that meaningfully reduce $\mathcal{J}$, rather than selecting a single K mechanically.

b) Stage 2 - Silhouette Analysis: For each candidate K identified in Stage 1, we compute the mean Silhouette score across all sellers. For seller i , the Silhouette coefficient is defined as:

$$s(i) = \frac{b(i) - a(i)}{\max\{a(i), b(i)\}} \quad (37)$$

where $a(i)$ is the mean intra-cluster distance from seller i to all other members of its assigned cluster, and $b(i)$ is the mean distance from seller i to all sellers in the nearest neighboring cluster:

$$a(i) = \frac{1}{|\mathcal{C}_{k(i)}| - 1} \sum_{j \in \mathcal{C}_{k(i)}, j \neq i} \|\mathbf{x}_i - \mathbf{x}_j\|$$

$$b(i) = \min_{k \neq k(i)} \frac{1}{|\mathcal{C}_k|} \sum_{j \in \mathcal{C}_k} \|\mathbf{x}_i - \mathbf{x}_j\| \quad (38)$$

The coefficient $s(i) \in [-1, 1]$, where values close to +1 indicate that seller i is well-matched to its own cluster and clearly separated from neighboring clusters, values near 0 indicate boundary membership, and negative values suggest possible misassignment. The mean Silhouette score over the full population:

$$\bar{s}(K) = \frac{1}{N} \sum_{i=1}^N s(i) \quad (39)$$

is used as the primary selection criterion: the value of K that maximizes $\bar{s}(K)$ within the candidate range identified by the elbow chart is chosen as the final number of clusters.

c) Final Configuration: With the optimal number of clusters K being selected based on the two-stage procedure described above, the final K-Means model is fitted on $\mathbf{X}_{\text{scaled}}$ with `n_init` = 20 random initializations and `max_iter` = 300 iterations per run, using `random_state` = 42 for reproducibility. The solution with the lowest inertia across all initializations is retained as the final clustering.

3) Cluster Interpretation:

a) Cluster Profiling: Once the final K-Means model is fitted on $\mathbf{X}_{\text{scaled}}$, each seller is assigned a cluster label $k(i) \in \{0, 1, \dots, K-1\}$. Cluster interpretation is performed by computing, for each cluster k , the mean and median of the original (untransformed) business-scale features: $\hat{y}_i^{(t^*)}$, $\hat{p}_i^{(t^*)}$, and $\hat{c}_i^{(t^*)}$. These statistics are computed on the raw feature values rather than the scaled inputs, so that the resulting profiles remain directly interpretable in business terms.

A key diagnostic of the segmentation outcome is the disparity between each cluster's seller share and its corresponding commission share. A cluster whose commission share substantially exceeds its seller share indicates a revenue-concentrated segment, whereas the reverse pattern is consistent with a low-activity or dormant population. This concentration structure has direct implications for platform resource allocation and seller retention prioritization and is discussed in detail in the results chapter.

For clusters exhibiting moderate engagement probability alongside disproportionately low transaction frequency, the discrepancy between $\hat{p}_i^{(t^*)}$ and $\hat{y}_i^{(t^*)}$ is interpreted as evidence of platform-side friction rather than latent churn, since a truly churned seller would be expected to show near-zero values on both dimensions simultaneously. Conversely, clusters with near-zero values on all three features are interpreted as dormant, with the caveat that seasonal inactivity cannot be distinguished from permanent churn within the clustering framework alone.

b) *Visual Inspection*: The clustering solution was visualized in both the raw business feature space and the Yeo-Johnson scaled model space using three-dimensional scatter plots.

Inspection of the raw space reveals the characteristic right-skewed distribution of commission and order volumes and confirms the necessity of the power transformation prior to clustering, as Euclidean distance in the raw space would be dominated by commission magnitude, effectively reducing the clustering to a univariate partition.

In the scaled space, the three clusters arrange along a diagonal gradient from low to high across all features simultaneously, consistent with the high inter-feature correlation structure of the data. This linear arrangement suggests that the three input features share a common latent dimension interpretable as overall seller activity level.

V. RESULTS

A. Data Assumption Validation

Prior to model fitting, a structured series of assumption checks is conducted on the calibration dataset comprising 2,242 sellers, spanning January 2017 through April 2018. The checks are organized into two groups corresponding to the BG/NBD and Gamma-Gamma models respectively.

1) BG/NBD Assumption Checks:

a) *Poisson Transaction Process*: The Poisson process assumption requires that inter-order times follow an exponential distribution, which implies a coefficient of variation (CV) of inter-order gaps equal to approximately 1.0 at the individual seller level.

The analysis was restricted to sellers with at least five repeat transactions ($N = 1,213$ sellers) to ensure reliable estimation of inter-order gap variability. The median CV of inter-order gaps is 1.16 and the mean CV is 1.25, both in close proximity to the theoretical value of 1.0. The P10-P90 range spans 0.79 to 1.81, indicating that the assumption is broadly satisfied across the seller population. While 49.2% of inter-order gaps are shorter than one day, this reflects genuine multi-order behavior within a marketplace setting rather than a data artifact, as transactions have already been consolidated at the seller-order level.

b) *Gamma Heterogeneity in Transaction Rates and Beta Heterogeneity in Dropout Probabilities*: The Gamma mixing distribution requires that transaction rates exhibit substantial heterogeneity across sellers, typically manifested as a strongly right-skewed frequency distribution. The calibration sample of 2,242 sellers displays exactly this pattern: the mean frequency of 30.9 orders far exceeds the median of 6, with the distribution

extending to a maximum of 1,515 (Table II). The concentration analysis provides further corroborating evidence: the top 1% of sellers (22 sellers) account for 24.3% of total orders, while the top 10% account for 66.5% (Table III), consistent with a Pareto-like dispersion arising from Gamma-distributed transaction rates rather than domination by a small group of extreme outliers.

TABLE II
SELLER REPEAT ORDER FREQUENCY DISTRIBUTION (CALIBRATION PERIOD)

Statistic	Value
Total sellers	2,242
Repeat sellers ($x > 0$)	1,828 (81.5%)
One-time sellers	414 (18.5%)
Mean frequency	30.9
Median frequency	6
P90 frequency	70
P99 frequency	398
Max frequency	1,515

TABLE III
SELLER TRANSACTION CONCENTRATION (CALIBRATION PERIOD)

Top %	N Sellers	Order Share	GMV Share
1%	22	24.3%	24.6%
5%	112	51.4%	51.5%
10%	224	66.5%	65.7%

In addition, according to Table II, the co-existence of a large group of low-frequency sellers (median = 6, $P25 = 1$) alongside a long-tailed group of high-frequency sellers ($P99 = 398$, $\max = 1,515$) also implies that dropout probabilities are widely dispersed across the population, consistent with a Beta distribution over $(0, 1)$ rather than a degenerate point mass. Sellers with high estimated dropout probability contribute predominantly to the lower tail of the frequency distribution, while those with low dropout probability populate the upper tail.

c) *Geometric Dropout Process*: The geometric dropout assumption states that sellers can permanently cease activity immediately after any transaction, with a constant per-transaction dropout probability p . Although the assumption cannot be tested directly from aggregate statistics, the observed proportion of one-time sellers provides empirical evidence consistent with the underlying mechanism. Specifically, 18.5% of sellers completed only a single transaction during the calibration period, while 81.5% generated at least one repeat order. The presence of a substantial one-time seller group is consistent with the geometric dropout mechanism, under which some sellers disengage immediately after their first transaction with the platform.

d) *Independence of λ and p* : The independence between transaction rate and dropout probability is a structural assumption of the BG/NBD model that cannot be directly tested from observed data alone. Its validity is evaluated indirectly through the stationarity check. As reported in the stationarity check below, the median late-to-early order rate ratio of 1.39 is driven primarily by marketplace-level growth rather than a selective

pattern in which high-rate sellers also exhibit systematically different dropout behavior, providing no evidence against the independence assumption.

Stationarity of Transaction Rates The BG/NBD model further requires that each seller’s transaction rate λ remain approximately constant over time. Marketplace-level diagnostics reveal a clear growth trend: orders per active seller increased from approximately 3.4 in January 2017 to 7.5 in January 2018, a approximately two-time increase over the calibration window. At the individual seller level, the median late-to-early order rate ratio is 1.39, indicating a mild but non-negligible upward drift. This constitutes a soft violation of the stationarity assumption attributable to platform-wide buyer growth rather than intrinsic seller behavior change. The resulting bias is expected to manifest as a slight underestimation of future order volumes, and is considered manageable given the 16-month calibration window over which the model learns a time-averaged transaction rate.

2) Gamma-Gamma Model Assumptions:

a) Gamma-Distributed Transaction Values: The Gamma distributional assumption requires that monetary values be strictly positive and that the cross-sectional distribution of seller-level AOV be right-skewed, with most sellers concentrated at lower values and a long upper tail. Both conditions are satisfied in the calibration data. No zero or negative monetary values are present among the eligible sellers. The AOV distribution exhibits a skewness of 10.58 and a median of BRL 106.38 substantially below the mean of BRL 190.60, consistent with the expected Gamma shape (Table IV). Figure 4 further illustrates the strongly right-skewed distribution of seller-level AOV values.

TABLE IV
SELLER AVERAGE ORDER VALUE DISTRIBUTION (CALIBRATION PERIOD,
 $N = 1,828$)

Statistic	Value (BRL)
Min	3.70
P25	61.05
Median	106.38
Mean	190.60
P75	189.17
P95	640.52
Max	8,370.00
Std	340.70
Skewness	10.58
Excess kurtosis	201.73
Non-positive values	0

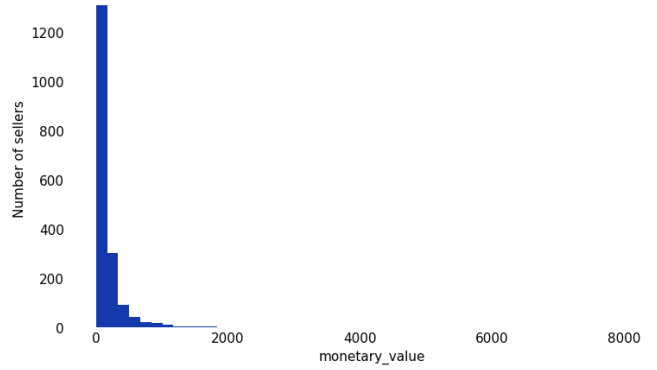


Fig. 4. Distribution of seller-level AOV in the calibration period.

b) Gamma Heterogeneity in Mean Transaction Values:

The Gamma mixing distribution on latent mean AOV requires that seller-level spending levels vary substantially across the population which is a condition clearly satisfied by the distributional statistics above. The ratio of standard deviation (340.70) to mean (190.60) exceeds 1.78, and the P95-to-median ratio of approximately 6.0 confirms the degree of cross-sectional dispersion in seller spending that the inverse-Gamma marginal distribution is designed to capture. The 81.5% model eligibility rate further ensures that parameter estimation is conducted on a large and representative sample rather than a select high-frequency subgroup.

c) Independence of Monetary Value and Transaction Process: The independence assumption is assessed through four correlation measures between seller repeat frequency and average order value, on both raw and log-transformed scales (Table V). All four metrics fall well below established thresholds, with the Spearman correlation being effectively zero at +0.005 on both scales. The extended Spearman correlation matrix confirms that monetary value is orthogonal to all R-F-T dimensions: its correlations with recency (+0.054) and tenure (−0.003) are negligible. These results provide exceptionally strong empirical support for the independence assumption.

TABLE V
FREQUENCY-MONETARY VALUE INDEPENDENCE CHECK

Scale	Metric	Value	Threshold
Raw	Pearson	−0.062	$ r < 0.15$
Raw	Spearman	+0.005	$ r < 0.15$
Log-log	Pearson	−0.033	$ r < 0.20$
Log-log	Spearman	+0.005	$ r < 0.20$

Stationarity of Monetary Value The temporal stability of seller AOV is evaluated on the 655 sellers with at least two transactions in both the early (before September 2017) and late (after September 2017) subperiods. The Pearson correlation between early-period and late-period AOV is 0.76, confirming that seller-level AOV rankings are well preserved over time (Figure 5).

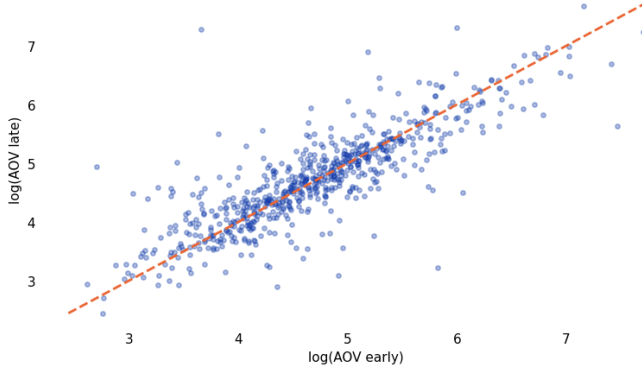


Fig. 5. Scatter plot of $\log(\text{AOV})$ early vs. late periods

B. Model Performance Assessment

1) Evaluation of the BG/NBD Transaction and Activity Model:

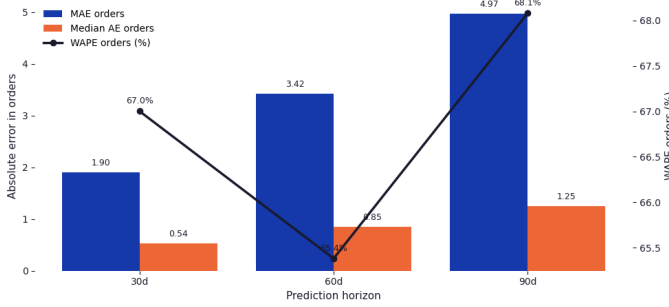


Fig. 6. Error metrics across 3 horizons of BG/NBD model

a) *Order Volume Accuracy:* Figure 6 shows that absolute error increases monotonically with horizon length, with MAE rising from 1.90 at 30 days to 3.42 at 60 days and 4.97 at 90 days. This is an expected pattern, as cumulative order counts grow with the length of the prediction window. More informatively, WAPE remains relatively stable across horizons, ranging from 65.4% to 67.8%, indicating that the proportional error at the seller level does not deteriorate substantially as the horizon extends. The persistently elevated WAPE reflects the intrinsic difficulty of point-forecasting individual seller order volumes in a setting characterized by high behavioral heterogeneity.

b) *Seller Activity Classification:* The model demonstrates stronger performance in distinguishing active from inactive sellers than in absolute order count forecasting. PR-AUC improves from 0.865 at 30 days to 0.906 at 60 days and 0.915 at 90 days, while ROC-AUC remains stable at approximately 0.90 across all horizons (Figure 7), indicating robust and consistent ranking of sellers by their activity probability.

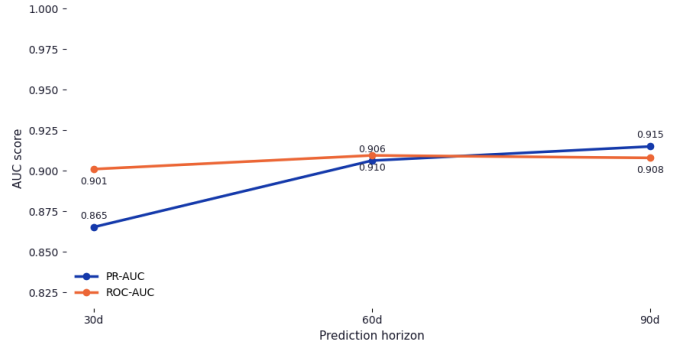


Fig. 7. Active Seller Classification Quality: PR-AUC vs. ROC-AUC

The lift analysis further confirms this: Lift@10 of $2.34\times$ at 30 days indicates that concentrating attention on the top-decile sellers ranked by predicted order volume yields an active seller density 2.34 times higher than random selection. Lift@10 declines to $2.00\times$ and $1.87\times$ at 60 and 90 days respectively (Figure 8), a natural consequence of a larger proportion of sellers becoming active over longer horizons, which reduces the discriminative advantage of targeting the top decile.

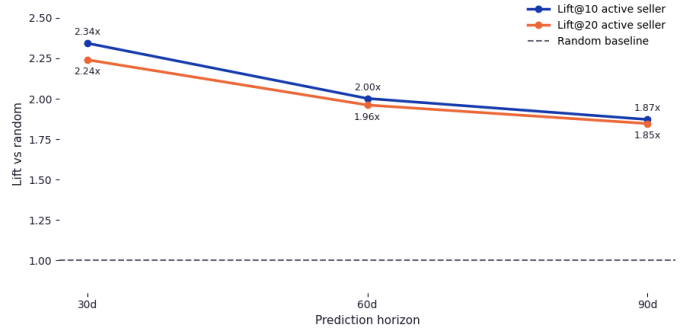


Fig. 8. Active Seller Targeting Lift Across Horizons

These results suggest that the BG/NBD model is best suited for seller activity ranking and prioritization rather than precise individual-level order count forecasting.

2) Evaluation of the Gamma-Gamma Monetary Model:

a) *AOV Point Estimation Accuracy:* In contrast to BG/NBD, Gamma-Gamma metrics improve as the horizon extends. WAPE on AOV decreases from 43.3% at 30 days to 41.6% at 60 days and 38.3% at 90 days, and MAE declines from 75.98 to 72.05 over the same range (Figure 9). This improvement occurs because longer holdout windows contain more transactions per seller, causing the observed AOV to converge toward the seller's true latent mean which is the quantity the Gamma-Gamma model is designed to estimate. The persistent gap between MAE and Median AE across all horizons reflects the right-skewed AOV distribution: a small segment of sellers with unusually high-value orders inflates the mean absolute error, and precise point estimation for this segment should not be expected.

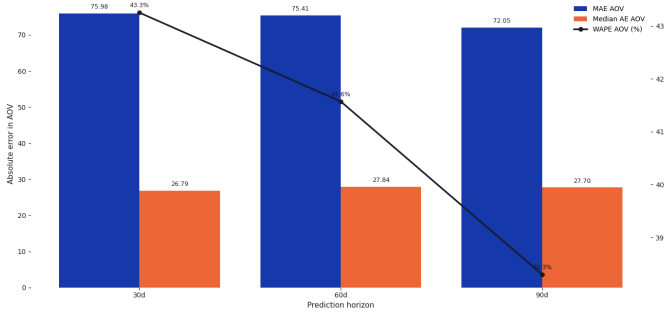


Fig. 9. Active Seller Targeting Lift Across Horizons

b) AOV Ranking Quality: The more operationally relevant capability of the Gamma-Gamma model lies in its ability to rank sellers by expected monetary value. Spearman rank correlation between predicted and actual AOV is stable and high across all horizons, ranging from 0.744 at 30 days to 0.769 at 90 days (Figure 12). This indicates that the model reliably preserves the relative ordering of sellers by spending level, even where absolute estimates are imprecise. For a marketplace operator, this ranking capability is of direct strategic value: it enables the platform to identify which sellers generate disproportionate per-order revenue independently of their transaction frequency.

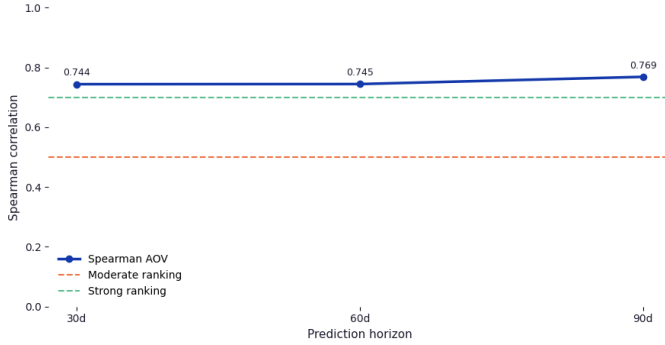


Fig. 10. Active Seller Targeting Lift Across Horizons

3) Integrated SeLTV Estimation Performance: Table VI reports the evaluation metrics for the combined BG/NBD and Gamma-Gamma pipeline on the expected commission target.

TABLE VI
COMBINED MODEL (EXPECTED COMMISSION) EVALUATION METRICS

Metric	30 days	60 days	90 days
MAE (commission)	44.93	77.36	111.32
Median AE (commission)	10.22	17.66	26.64
WAPE (commission)	0.753	0.708	0.737
Spearman (commission)	0.676	0.728	0.740

When the BG/NBD and Gamma-Gamma outputs are combined into a single expected commission estimate, the resulting metric directly reflects the platform’s revenue perspective, since marketplace income is structurally proportional to seller-generated GMV. MAE on commission increases from BRL 44.93 at 30 days to BRL 77.36 and BRL 111.32 at 60

and 90 days respectively, consistent with the accumulation of absolute error over longer windows. WAPE on commission ranges from 70.8% to 75.3%, indicating that precise seller-level commission forecasting remains challenging. This is an outcome primarily attributable to order volume uncertainty from the BG/NBD component, as the Gamma-Gamma model demonstrates superior stability on its own.

The Spearman rank correlation on commission, however, presents a more favorable picture: it improves from 0.676 at 30 days to 0.728 at 60 days and 0.740 at 90 days. Surpassing 0.7 at the 60-day horizon is a meaningful threshold, as it indicates that the combined model produces a reliable seller ranking by expected revenue contribution. From a business operations standpoint, decisions such as seller prioritization, support resource allocation, and engagement campaign targeting do not require exact commission forecasts; they require a trustworthy ordering of sellers by value potential. The Spearman results demonstrate that the combined model meets this requirement with increasing reliability as the horizon extends.

4) Comparative Analysis of Prediction Horizons: The evaluation results across the three horizons reveal complementary trade-offs, which are summarized in Table VII.

TABLE VII
HORIZON COMPARISON SUMMARY

Criterion	30 days	60 days	90 days
Order volume WAPE	0.670	0.654	0.678
PR-AUC	0.865	0.906	0.915
Lift@10%	2.34×	2.00×	1.87×
WAPE AOV	0.433	0.416	0.383
Spearman AOV	0.744	0.745	0.769
WAPE commission	0.753	0.708	0.737
Spearman commission	0.676	0.728	0.740

The 30-day horizon offers the strongest short-term activity discrimination, with Lift@10 of 2.34×, making it most appropriate for near-term seller activation campaigns or supply forecasting. However, Spearman commission rank correlation at 30 days is only 0.676, the lowest across all horizons, indicating that the commission ranking signal contains more noise at this scale potentially because sellers with high long-term value may not generate sufficient order volume within a single month.

The 90-day horizon produces the highest Spearman scores for both AOV (0.769) and commission (0.740), confirming its suitability for strategic seller tier identification over the medium term. However, this horizon also produces the highest absolute errors and exhibits a tendency toward optimistic commission forecasts due to error accumulation.

The 60-day horizon achieves the best balance across all dimensions simultaneously. It produces the lowest WAPE on order volume (65.4%), strong classification performance (PR-AUC = 0.906, Lift@10 = 2.00×), the lowest WAPE on commission (70.8%), and a Spearman commission rank correlation of 0.728, which crosses the operational relevance threshold of 0.7. Critically, 60 days is long enough to reduce the short-term behavioral noise that limits the 30-day commission signal, yet short enough to prevent the error accumulation that reduces forecast reliability at 90 days.

On this basis, the 60-day prediction horizon is selected as the input to the K-Means segmentation model. The three clustering features including $\hat{y}_i^{(60)}$, $\hat{p}_i^{(60)}$, and $\hat{c}_i^{(60)}$ are derived from this horizon, representing the best available balance between predictive accuracy, seller ranking reliability, and operational actionability.

5) *K-means:*

a) *Selecting the Optimal Number of Clusters:* To determine the optimal number of clusters for the K-Means model, the study applied the Elbow Method and Silhouette Score. The Elbow Method was used to observe how the Within-Cluster Sum of Squares (WCSS) changes as the number of clusters increases from 2 to 10. WCSS measures the total squared distance between each observation and the centroid of its assigned cluster. A lower WCSS indicates that observations within the same cluster are more compact.

The result shows that WCSS decreases substantially as k increases. Specifically, WCSS drops from 1,582.45 at $k=2$ to 945.80 at $k=3$, and then to 643.30 at $k=4$. This indicates that moving from two to three clusters significantly improves cluster compactness, while moving from three to four clusters also provides additional improvement. However, after $k=4$, the WCSS curve starts to flatten, suggesting that adding more clusters produces smaller marginal gains. Based on this pattern, $k=3$ and $k=4$ were identified as the main elbow candidates.

Although $k=4$ has a lower WCSS than $k=3$, this alone is not sufficient to justify selecting $k=4$, because WCSS naturally decreases whenever the number of clusters increases. Therefore, Silhouette Score was used as an additional validation metric to compare the quality of the two candidate solutions. The Silhouette Score evaluates both intra-cluster cohesion and inter-cluster separation. A higher Silhouette Score indicates that observations are better matched to their own cluster and more clearly separated from other clusters.

The comparison shows that $k=3$ achieves a Silhouette Score of 0.5295, while $k=4$ achieves a lower score of 0.4789. This means that the three-cluster solution provides better separation and more coherent cluster structure than the four-cluster solution. Although $k=4$ reduces WCSS further, the lower Silhouette Score suggests that the additional cluster may fragment the data without improving the overall cluster quality.

Therefore, $k=3$ was selected as the final number of clusters. This choice provides a good balance between statistical quality and business interpretability. From a managerial perspective, three clusters are also easier to translate into actionable seller segments: Dormant Sellers, Mid-Tier Engaged Sellers, and High-Value Active Sellers. This segmentation structure is simple enough for business decision-making while still capturing meaningful differences in seller activity and expected financial value.

b) *Internal clustering metrics:* The K-Means solution applied to the Yeo-Johnson transformed feature space yielded internal validity scores that collectively indicate a geometrically coherent and well-separated partition.

The Silhouette Score of 0.5295 falls at the boundary conventionally considered acceptable, where values exceeding

0.50 denote meaningful cluster structure (Rousseeuw, 1987).

c) *Cluster Profile:* The K-Means clustering framework successfully identified three distinct seller segments characterized by substantial differences in transactional behavior, activity probability, and projected commercial contribution. By integrating probabilistic outputs derived from the BG/NBD and Gamma-Gamma models, specifically expected future orders, estimated seller survival probability, and projected commission value over a 60-day horizon, the proposed framework provides a behaviorally grounded approach to seller segmentation beyond traditional descriptive clustering techniques.

Table belows summarizes the key descriptive statistics for each cluster alongside each segment's proportional contribution to total seller count and projected platform commission over the 60-day forecasting horizon.

TABLE VIII
CLUSTER SEGMENTATION SUMMARY (60-DAY HORIZON)

Criterion	$k = 2$	$k = 1$	$k = 0$
Segment Label	High-Value Active	Mid-Tier Engaged	Dormant
Number of Sellers	857	552	839
Seller Share	38.1%	24.6%	37.3%
Total Commission	243,000	20,500	3,050
Commission Share	91.1%	7.7%	1.1%

Cluster 2: High-Value Active Sellers

The K-Means clustering model identified Cluster 2 as the platform's most commercially valuable seller segment. This cluster is characterized by high expected future transaction activity, near-certain purchase probability, and the largest projected commission contribution within the next 60 days.

TABLE IX
CLUSTER 2 (HIGH-VALUE ACTIVE SELLERS) PROFILE SUMMARY

Metric	Mean	Median
Expected orders (60d)	13.88 orders	6.81 orders
P(purchases)	98.6%	99.9%
Expected commission (60d)	283.50	135.97

Sellers in this cluster exhibit extremely strong future activity signals. The median purchase probability reaches approximately 99.9%, indicating that these sellers are behaviorally active and show minimal signs of churn or declining engagement under the BG/NBD framework. In addition, Cluster 2 records the highest expected order frequency among all seller groups, confirming its role as the platform's core active seller base.

The projected commission value is also substantially higher than that of other clusters. However, the large gap between mean and median values for both expected orders and expected commission suggests a strongly right-skewed distribution. This indicates that a small number of top-tier sellers contribute disproportionately high transaction volumes and commission revenue, reflecting a long-tail value structure commonly observed in marketplace ecosystems.

From a business perspective, Cluster 2 functions as the platform's primary revenue engine. Even a moderate increase in churn within this segment could therefore have a significant

impact on short-term commission performance. The internal heterogeneity of this cluster also suggests that further sub-segmentation may be useful to distinguish ultra-high-value sellers from moderately high-performing active sellers.

Strategically, Cluster 2 should receive the highest operational priority. Recommended actions include retention-focused seller relationship management, loyalty programs, premium visibility support, and early-warning systems for churn detection.

Cluster 1: Mid-Tier Engaged Sellers

Cluster 1 represents a transitional seller segment with moderate engagement and limited but meaningful projected commercial contribution. Sellers in this group are not fully inactive, but they generate considerably lower expected order volume and commission than Cluster 2.

TABLE X
CLUSTER 1 (MID-TIER ENGAGED SELLERS) PROFILE SUMMARY

Metric	Mean	Median
Expected orders (60d)	1.41 orders	1.27 orders
P(purchases)	70.0%	71.8%
Expected commission (60d)	37.20	23.50

The median purchase probability of 71.8% suggests that most sellers in this cluster still have a reasonable likelihood of remaining active. However, their expected order frequency remains low, indicating that the main challenge is not complete disengagement but insufficient transaction conversion.

Compared with Cluster 2, Cluster 1 displays a more homogeneous behavioral profile, with less evidence of extreme outliers. Sellers in this segment continue to interact with the platform, yet their commercial contribution remains limited. This pattern may reflect operational friction, weak product discoverability, limited marketplace visibility, onboarding difficulties, seasonal demand fluctuations, or competition from alternative platforms.

From a strategic perspective, Cluster 1 represents the platform's strongest growth opportunity. Since activating existing sellers is generally more cost-effective than acquiring new sellers, this segment is well suited for targeted development initiatives. Recommended actions include onboarding optimization, personalized incentives, activation campaigns, seller coaching, and improved visibility support.

Cluster 0: Dormant Sellers

Cluster 0 consists of sellers with weak future activity signals and minimal projected commercial contribution. These sellers have the lowest expected order frequency, the lowest purchase probability, and the smallest expected commission value among the three clusters.

TABLE XI
CLUSTER 0 (DORMANT SELLERS) PROFILE SUMMARY

Metric	Mean	Median
Expected orders (60d)	0.16 orders	0.10 orders
P(purchases)	13.6%	9.7%
Expected commission (60d)	3.64	1.54

The BG/NBD model assigns very low survival probabilities to sellers in Cluster 0, with the median purchase probability

falling below 10%. This indicates that most sellers in this segment are behaviorally close to churn and are unlikely to generate meaningful transaction activity in the near future.

Both expected order volume and expected commission are nearly negligible. Although Cluster 0 represents a sizeable share of the seller base, its contribution to projected commission remains marginal, making it a low-value segment from a short-term revenue perspective.

The main business issue is therefore not direct revenue loss, but operational inefficiency and opportunity cost. Intensive resources allocated to this group may generate limited returns and could be more effectively redirected toward higher-potential sellers in Clusters 1 and 2.

However, sellers in Cluster 0 should not be automatically treated as permanently inactive, as some may be seasonal sellers. Therefore, this segment should be managed through lightweight and scalable interventions, such as automated reactivation campaigns, low-cost reminders, inactive seller monitoring, and periodic reassessment of seller status.

d) *Visual Inspection:* To complement the quantitative clustering metrics, the segmentation results are visually inspected in both the raw business feature space and the transformed model space.

Raw Business Space Visualization

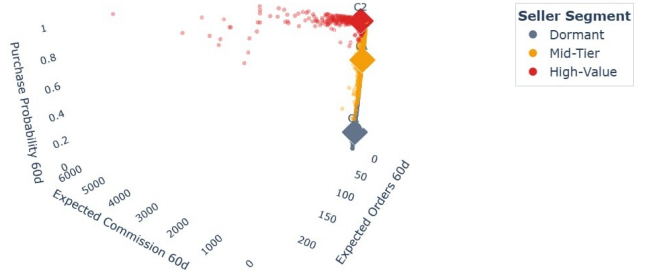


Fig. 11. Raw Business Feature Space

The first 3D chart visualizes the seller clusters in the original business space using three key predictive variables: expected commission, probability of at least one purchase, and expected orders within 60d. This representation is particularly useful from a managerial perspective because each axis preserves its original business interpretation: expected commission revenue, expected transaction volume, and the probability that a seller will generate future orders within the next 60 days.

In contrast, the Dormant Sellers cluster is concentrated in the low-value region of the feature space, characterized by low expected commission, low expected order volume, and low probability of future activity. This segment represents approximately 37.3% of sellers but contributes only around 1.1% of total expected commission. The result suggests that a large proportion of the seller base currently generates very limited short-term business value. Consequently, allocating intensive manual support or expensive retention programs to this segment may not be economically efficient.

The remaining group, referred to as Mid-Tier Engaged Sellers, lies between the high-value and dormant segments. Although these sellers do not yet generate the same level of expected value as the high-value cluster, they still exhibit substantially stronger activity signals than dormant sellers. This segment can therefore be interpreted as a potential growth opportunity for the platform. Appropriate strategies for this group may include seller education programs, listing optimization support, operational guidance, and targeted incentive mechanisms designed to help sellers transition toward the high-value segment.

From a managerial perspective, the clustering result demonstrates that seller management should not rely on a one-size-fits-all strategy. Instead, the platform should prioritize resource allocation toward high-value sellers, implement growth-oriented development programs for mid-tier sellers, and apply low-cost automated reactivation campaigns for dormant sellers.

Clustering in the Scaled Model Space

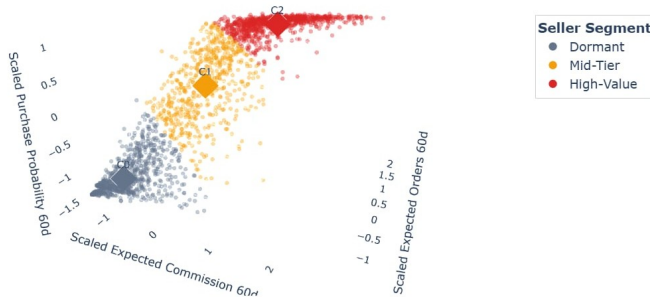


Fig. 12. Scaled K-Means Model Space

The second 3D chart presents the clustering result in the transformed and scaled model space. Unlike the raw business space chart, this visualization reflects the actual feature space used by the K-Means algorithm to calculate distances and assign cluster labels.

Scaling is an essential preprocessing step because the three input variables have very different ranges and distributions. Expected commission usually has a much larger scale and higher dispersion than purchase probability. Without scaling, K-Means could be dominated by the monetary variable, causing the clusters to mainly reflect differences in expected commission rather than a balanced combination of commission potential, expected transaction volume, and future activity probability.

In the scaled model space, the three clusters are separated in a more technically meaningful way. The Dormant Sellers cluster is positioned in the low-value region across all three standardized dimensions, indicating sellers with weak short-term potential. The High-Value Sellers cluster is located in the high-value region, representing sellers with strong expected activity, high expected orders, and high expected commission. The Mid-tier Sellers cluster lies between these two groups,

functioning as a transitional segment between inactive sellers and high-value sellers.

The key insight from this chart is that the clustering result is not based solely on absolute revenue potential. Instead, the model segments sellers based on a balanced combination of three dimensions: future activity probability, expected order volume, and expected commission value. This makes the segmentation more robust and reliable, as each seller is evaluated using both behavioral potential and financial potential.

6) *Marketing Strategy*: The clustering results provide a practical basis for designing differentiated seller management strategies. Instead of applying the same marketing approach to all sellers, Olist should allocate resources according to each segment's expected future value. This follows the logic of CLV-based marketing, where marketing investment is prioritized based on expected contribution rather than treating all customers, or in this case sellers, equally.

In this study, sellers are segmented using three forward-looking indicators: expected orders within 60 days, probability of future activity, and expected commission within 60 days. These variables allow Olist to design strategies based on predicted seller potential rather than historical performance alone. Based on the clustering results, three strategic approaches are proposed: Retain and Grow for High-Value Active Sellers, Activate and Develop for Mid-Tier Engaged Sellers, and Automate and Diagnose for Dormant Sellers.

a) *High-Value Active Sellers: Retain and Grow*: High-Value Active Sellers have high expected orders, high probability of future activity, and the largest expected commission contribution. This group is the main revenue driver for Olist. Therefore, the priority is not to activate them, but to retain them and help them grow further.

The recommended campaign is the VIP Seller Growth Program. Olist should provide this group with priority support, faster issue resolution, personalized performance reports, and recommendations on pricing, product listings, inventory planning, and catalog expansion. These actions help reduce operational friction and strengthen seller loyalty.

Olist should also invite these sellers to major promotional campaigns and co-marketing activities. Since they already have strong transaction potential, additional visibility is more likely to generate actual order growth. Suitable benefits include premium campaign access, advertising support, featured placement, and early access to new platform tools.

For this segment, incentives should focus on strategic support rather than simple discounts. Olist should avoid unnecessary margin loss and instead provide benefits that improve seller satisfaction and long-term commitment.

The main KPIs include seller retention rate, actual commission uplift, order growth, campaign participation rate, catalog expansion, and churn risk reduction.

b) *Mid-Tier Engaged Sellers: Activate and Develop*: Mid-Tier Engaged Sellers have moderate activity potential. They are not inactive, but their expected orders and expected commission remain much lower than those of high-value sellers. This group

should be treated as a growth opportunity because some sellers may develop into higher-value sellers if supported properly.

The recommended campaign is the Seller Acceleration Program. The objective is to improve seller capability and increase transaction activity. Olist can provide email guidance, short training modules, webinars, and checklists on product listing quality, pricing, product images, stock management, delivery reliability, campaign participation, and review management.

Support for this group should be semi-personalized. For example, sellers with few products can receive catalog expansion suggestions, while sellers with weak descriptions can receive listing improvement guidance. Sellers with irregular activity can receive reminders to update inventory or join suitable campaigns.

Olist may also offer small conditional incentives, such as listing support vouchers, limited advertising credits, or temporary visibility boosts. These incentives should be linked to clear actions, such as adding new products, improving listing completeness, updating inventory, or joining a promotional campaign.

The main KPIs include listing update rate, number of new products added, campaign participation rate, active probability uplift, actual order growth, commission uplift, and the share of sellers moving from Mid-Tier Engaged to High-Value Active.

c) Dormant Sellers: Automate and Diagnose: Dormant Sellers have low expected orders, low probability of future activity, and very low expected commission. Although this group represents a large share of the seller base, its expected revenue contribution is limited. Therefore, Olist should avoid expensive manual support for this segment.

The recommended campaign is Seller Reactivation Automation. The goal is to test reactivation potential at low cost and identify why sellers became inactive. Olist can use automated emails or in-platform notifications with a simple reactivation checklist, such as updating inventory, reviewing listings, checking prices, confirming delivery settings, and joining a low-barrier campaign.

Olist should also collect feedback through short surveys. These surveys can identify whether inactivity is caused by low demand, high shipping costs, platform fees, inventory issues, technical problems, or competition from other marketplaces. This helps Olist distinguish sellers with recovery potential from sellers unlikely to return.

Incentives for this segment should be small, temporary, and conditional. For example, Olist can offer limited listing support or short-term advertising credits only if the seller completes a reactivation action. Dedicated account managers should not be used unless a seller shows clear recovery potential.

The main KPIs include reactivation rate, cost per reactivated seller, email open rate, click-through rate, checklist completion rate, survey response rate, and post-reactivation commission.

VI. A/B TESTING DESIGN AND STATISTICAL EVALUATION FRAMEWORK

This study uses the Brazilian E-Commerce Public Dataset by Olist as the empirical context for designing seller-oriented,

data-driven marketing strategies. The dataset provides historical information on orders, sellers, products, payments, reviews, and delivery performance in a marketplace setting. Since the dataset is observational and does not contain an actual experimental intervention, the following section should be interpreted as a proposed A/B testing framework for future implementation rather than as a completed causal experiment.

After segmenting sellers based on three predictive variables, namely expected order volume, expected commission, and probability of being active within 60 days, three different marketing strategies are proposed for three seller segments. However, the effectiveness of these strategies should not be evaluated only through simple descriptive comparisons between the control and treatment groups. Instead, the study adopts a statistical testing framework to assess whether the observed differences are likely to reflect a real treatment effect or merely random variation.

In each experiment, the unit of randomization is the seller, not the order. This choice is important because marketing interventions are assigned at the seller level. If orders were treated as independent observations, the analysis could underestimate uncertainty because multiple orders may belong to the same seller. Therefore, each seller is assigned to only one of two groups: Control A, which receives the current or standard support, and Treatment B, which receives the new campaign.

Let i denote an individual seller and let D_i be the treatment assignment indicator:

$$D_i = \begin{cases} 0, & \text{if seller } i \text{ is assigned to Control A,} \\ 1, & \text{if seller } i \text{ is assigned to Treatment B.} \end{cases}$$

For each segment k , where

$$k \in \{1, 2, 3\},$$

the seller population in that segment is denoted by S_k . The three values of k correspond to High-value active sellers, Mid-tier engaged sellers, and Dormant sellers, respectively.

Since the number of sellers in each segment is not sufficiently large, the study proposes to conduct the experiment on the entire seller population within each segment rather than drawing a smaller sample. For each segment S_k , all sellers are randomly divided into a control group and a treatment group:

$$S_k = A_k \cup B_k,$$

where A_k is the control group and B_k is the treatment group. These two groups are mutually exclusive:

$$A_k \cap B_k = \emptyset.$$

A 50:50 split is recommended in order to maximize the comparability between the two groups:

$$n_{A_k} \approx n_{B_k}.$$

This design allows the experiment to use all available observations within each segment, thereby increasing statistical power and reducing the uncertainty caused by small sample size. This is particularly relevant in marketplace experimentation,

where some segments may contain a limited number of eligible participants [10].

The average treatment effect for segment k is estimated as the difference in the primary metric between Treatment B and Control A:

$$\hat{\tau}_k = \bar{Y}_{B_k} - \bar{Y}_{A_k},$$

where Y_i represents the primary metric of seller i , $\bar{Y}_{B_k}$ is the average value of the primary metric in Treatment B, and $\bar{Y}_{A_k}$ is the average value in Control A.

The statistical hypotheses are defined as follows:

$$H_0 : \tau_k = 0,$$

$$H_1 : \tau_k > 0.$$

The null hypothesis H_0 states that the new campaign does not improve the primary metric compared with the current support condition. The alternative hypothesis H_1 states that the new campaign generates a positive treatment effect.

For continuous primary metrics, such as actual net commission, the study uses an independent two-sample t -test. If equal variances between the two groups cannot be assumed, Welch's t -test is preferred because it does not require the equal-variance assumption [8]. The test statistic for Welch's t -test is:

$$t = \frac{\bar{Y}_B - \bar{Y}_A}{\sqrt{\frac{s_B^2}{n_B} + \frac{s_A^2}{n_A}}},$$

where $\bar{Y}_B$ and $\bar{Y}_A$ are the sample means of the treatment and control groups, s_B^2 and s_A^2 are the sample variances, and n_B and n_A are the sample sizes of the two groups.

If the equal-variance assumption is considered reasonable, the pooled two-sample t -test can be used:

$$t = \frac{\bar{Y}_B - \bar{Y}_A}{s_p \sqrt{\frac{1}{n_B} + \frac{1}{n_A}}},$$

where the pooled variance is:

$$s_p^2 = \frac{(n_B - 1)s_B^2 + (n_A - 1)s_A^2}{n_B + n_A - 2}.$$

However, because marketplace sellers often differ substantially in order volume, revenue contribution, and commission generation, Welch's t -test is generally more appropriate in this context.

For proportion-based primary metrics, such as active rate and reactivation rate, the study uses a two-proportion z -test. This test evaluates whether the difference between two independent proportions is statistically significant [9]. The hypotheses are:

$$H_0 : p_B - p_A = 0,$$

$$H_1 : p_B - p_A > 0,$$

where p_B is the proportion of sellers achieving the target outcome in Treatment B and p_A is the corresponding proportion in Control A.

The z -statistic is calculated as:

$$z = \frac{\hat{p}_B - \hat{p}_A}{\sqrt{\hat{p}(1 - \hat{p}) \left(\frac{1}{n_B} + \frac{1}{n_A} \right)}},$$

where the pooled proportion is:

$$\hat{p} = \frac{x_A + x_B}{n_A + n_B}.$$

Here, x_A and x_B represent the number of sellers achieving the target outcome in the control and treatment groups, respectively.

The significance level is set at $\alpha = 0.05$.

If the resulting p -value is smaller than 0.05, the null hypothesis is rejected, suggesting that the campaign has a statistically significant positive effect. Nevertheless, statistical significance alone is not sufficient for business decision-making. A campaign should only be scaled if the estimated effect is also economically meaningful and if the guardrail metrics remain within acceptable risk thresholds.

A. Segment 1: High-value Active Sellers

The first segment consists of High-value active sellers. The proposed experiment for this segment is called the *VIP Seller Growth Program*. In Control A, sellers continue to receive Olist's standard support. In Treatment B, sellers receive VIP support, including account assistance, listing optimization advice, campaign support, performance reporting, and recommendations for catalog expansion.

The primary metric for this experiment is actual net commission within 60 days. Let c_i denote the actual commission generated by seller i , and let s_i denote the support cost associated with that seller. The actual net commission is defined as:

$$NC_i = c_i - s_i.$$

The actual commission can be written as:

$$c_i = \sum_{j=1}^{n_i} commission_{ij},$$

where n_i is the number of orders generated by seller i , and $commission_{ij}$ is the commission obtained from order j .

The treatment effect is estimated as:

$$\hat{\tau}_{NC} = \overline{NC}_B - \overline{NC}_A.$$

The hypotheses are:

$$H_0 : \overline{NC}_B - \overline{NC}_A = 0,$$

$$H_1 : \overline{NC}_B - \overline{NC}_A > 0.$$

Because actual net commission is a continuous variable, Welch's t -test is used to determine whether the treatment group generates significantly higher net commission than the control group. The campaign should be considered successful only if the estimated treatment effect is positive, statistically significant, and economically meaningful.

The economic condition can be expressed as:

$$\Delta c > \Delta s,$$

or equivalently:

$$\Delta NC > 0.$$

This means that the additional commission generated by the VIP program must exceed the additional support cost. Therefore, the VIP Seller Growth Program should only be scaled if it increases net commission after accounting for the cost of providing premium support.

Secondary metrics include actual order volume, number of newly added products, number of campaigns joined, and 60-day seller retention. These metrics provide additional insight into whether VIP support encourages sellers to expand their catalog and participate more actively in platform campaigns.

Guardrail metrics include cancellation rate, late delivery rate, refund rate, complaints, and support cost. For example, the cancellation rate is defined as:

$$\text{Cancellation Rate} = \frac{\text{Number of cancelled orders}}{\text{Total number of orders}}.$$

Even if the primary metric improves, the campaign should not be scaled if the treatment group shows a substantial deterioration in operational quality or if the support cost becomes excessive.

B. Segment 2: Medium-potential Sellers

The second segment consists of Medium-potential sellers. The proposed experiment for this segment is called the *Seller Acceleration Program*. In Control A, sellers receive general emails or standard notifications. In Treatment B, sellers receive a structured sequence of educational emails, webinars, store optimization checklists, product or inventory update suggestions, and small campaign incentives.

The primary metric is the actual active rate within 60 days. Let A_i be a binary variable indicating whether seller i becomes active:

$$A_i = \begin{cases} 1, & \text{if seller } i \text{ has at least one order within 60 days,} \\ 0, & \text{otherwise.} \end{cases}$$

The active rate is calculated as:

$$AR = \frac{\sum_{i=1}^N A_i}{N},$$

where N is the total number of sellers in the group.

The treatment effect is:

$$\hat{\tau}_{AR} = AR_B - AR_A.$$

The hypotheses are:

$$H_0 : AR_B - AR_A = 0,$$

$$H_1 : AR_B - AR_A > 0.$$

Since the primary metric is a proportion, a two-proportion z -test is used. If the p -value is below 0.05 and $AR_B > AR_A$, the Seller Acceleration Program can be considered statistically effective in increasing seller activity.

Secondary metrics include listing update rate, number of new products, actual order volume, actual commission, and upgrade rate to the high-value segment. The upgrade rate is defined as:

$$\text{Upgrade Rate} = \frac{\text{Number of sellers moving}}{\text{Total number of medium-potential sellers}}.$$

Guardrail metrics include unsubscribe rate, spam complaints, voucher cost, and cancellation rate. These metrics are especially important because the treatment involves more frequent communication and promotional incentives. If sellers perceive the intervention as excessive or irrelevant, unsubscribe and spam complaint rates may increase.

The main evaluation window is 60 days. However, early indicators should also be monitored on day 14 and day 28. These early signals do not replace the final statistical test but can help detect whether the campaign is moving in the expected direction. The campaign should only be rolled out if the active rate improves significantly, the cost remains reasonable, and the guardrail metrics remain within acceptable limits.

C. Segment 3: Low-value and Low-active Sellers

The third segment consists of Low-value or Low-active sellers. These sellers have low expected order volume, low expected commission, and low probability of being active within 60 days. Because this group has relatively low expected economic value, Olist should avoid investing substantial manual support resources. The appropriate strategy is *Automate and Diagnose*. The objective is to use low-cost automated communication to reactivate sellers while collecting information about why they became inactive.

The proposed experiment for this segment is called *Seller Reactivation Automation*. In Control A, sellers receive no special campaign or only general newsletters. In Treatment B, sellers receive automated emails, a checklist for returning to selling, a survey asking for reasons for inactivity, and a small time-limited incentive.

The primary metric is the reactivation rate within 60 days. Let R_i be a binary variable indicating whether seller i is reactivated:

$$R_i = \begin{cases} 1, & \text{if seller } i \text{ becomes active again within 60 days,} \\ 0, & \text{otherwise.} \end{cases}$$

The reactivation rate is:

$$RR = \frac{\sum_{i=1}^N R_i}{N}.$$

The treatment effect is:

$$\hat{\tau}_{RR} = RR_B - RR_A.$$

The hypotheses are:

$$H_0 : RR_B - RR_A = 0,$$

$$H_1 : RR_B - RR_A > 0.$$

Because reactivation rate is a proportion, a two-proportion z -test is used to evaluate whether Treatment B significantly increases the probability of reactivation compared with Control A.

However, for this low-value segment, statistical significance must be interpreted together with cost efficiency. Let $C_{campaign}$ denote the total campaign cost and let $N_{reactivated}$ denote the number of sellers reactivated by the campaign. The cost per reactivated seller is:

$$CPR = \frac{C_{campaign}}{N_{reactivated}}.$$

The campaign should only be maintained if:

$$CPR < \bar{c}_{reactivated},$$

where $\bar{c}_{reactivated}$ is the average commission generated by reactivated sellers after returning to activity.

Secondary metrics include email open rate, click rate, survey response rate, actual orders, and actual commission. The email open rate and click rate are defined as:

$$\begin{aligned} \text{Open Rate} &= \frac{\text{Number of opened emails}}{\text{Number of delivered emails}}, \\ \text{Click Rate} &= \frac{\text{Number of clicked emails}}{\text{Number of opened emails}}. \end{aligned}$$

Guardrail metrics include spam complaints, unsubscribe rate, voucher abuse, and cost per reactivated seller. If the campaign leads to high spam complaints or excessive voucher abuse, it should be revised or discontinued even if the reactivation rate improves.

D. Guardrail Metrics and Rollout Decision Rules

In addition to primary and secondary metrics, guardrail metrics are included to control operational and business risks. Prior research on A/B testing with multiple metrics emphasizes that not all metrics serve the same purpose. Some metrics measure success, while others function as guardrails to prevent harmful side effects [11].

Let G_i denote a guardrail metric, such as cancellation rate, refund rate, late delivery rate, unsubscribe rate, or spam complaint rate. The difference between the treatment and control groups is:

$$\Delta G = \bar{G}_B - \bar{G}_A.$$

If a guardrail metric is undesirable when it increases, Olist should define a maximum acceptable deterioration threshold, denoted by δ_G . The campaign is considered safe only if:

$$\text{Upper } CI_{95\%}(\Delta G) < \delta_G.$$

For example, if Olist allows the cancellation rate to increase by at most one percentage point, then:

$$\delta_G = 0.01.$$

In this case, the campaign should only be rolled out if the upper bound of the 95% confidence interval for the increase in cancellation rate is below 0.01.

Therefore, a campaign is recommended for full rollout only if it satisfies the following three conditions:

- The primary metric improves significantly.
- The campaign generates positive economic value, meaning that $\text{Benefit}_{incremental} - \text{Cost}_{incremental} > 0$.
- The guardrail metrics remain within acceptable risk thresholds, meaning that $\text{Upper } CI_{95\%}(\Delta G) < \delta_G$.

The first condition ensures statistical validity, the second ensures economic value, and the third ensures that the campaign does not create unacceptable operational risks.

VII. CONCLUSION AND LIMITATIONS

A. Conclusion

This study demonstrates the feasibility of repurposing established buyer-side probabilistic models for seller-side lifetime value estimation in a B2B marketplace context. By applying the BG/NBD and Gamma-Gamma models to the Olist seller population, we show that the structural assumptions underlying these frameworks are sufficiently satisfied at the seller level to support meaningful probabilistic inference.

Regarding model performance, the BG/NBD and Gamma-Gamma models do not achieve exceptional point estimation accuracy. However, both models demonstrate substantially stronger performance on ranking tasks, confirming its reliability in distinguishing active from inactive sellers. These results suggest that the primary operational value of the framework lies in seller prioritization and relative value identification rather than precise revenue forecasting.

The comparative analysis of prediction horizons reveals complementary trade-offs. The 30-day horizon offers the strongest near-term activity discrimination, making it well suited for immediate activation campaigns. The 90-day horizon produces the highest Spearman scores for both AOV and commission ranking, making it appropriate for strategic medium-term tier classification. The 60-day horizon, however, achieves the best overall balance: it records the lowest WAPE on both order volume and commission and have a strong classification performance. The 60-day horizon is therefore selected as the basis for downstream segmentation.

The K-Means clustering model, applied to probabilistic features derived from the 60-day horizon, successfully partitions the seller population into three behaviorally and commercially distinct segments: High-Value Active Sellers, Mid-Tier Engaged Sellers, and Dormant Sellers. Each segment receives a differentiated strategy: a *Retain & Grow* approach via a VIP Seller Growth Program for high-value sellers; an *Activate & Develop* approach via a Seller Acceleration Program aimed at transitioning mid-tier sellers upward; and an *Automate & Diagnose* approach via low-cost reactivation automation for dormant sellers. These cluster assignments further serve as the behavioral grouping structure for cohort-specific A/B testing protocols, enabling causal evaluation of each targeted intervention.

B. Limitations and Future Developments

Despite these contributions, the study is subject to several limitations that warrant acknowledgment.

First, as demonstrated throughout the evaluation, both the BG/NBD and Gamma-Gamma models are better suited for seller ranking than for point forecasting. Absolute commission predictions carry substantial error, particularly at the individual seller level, and should be interpreted as relative value signals rather than precise revenue projections.

Second, the BG/NBD model exhibits a soft violation of the stationarity assumption. Transaction rates show a clear growth trend over the calibration window, driven by platform-wide buyer expansion rather than intrinsic seller behavior change. As a consequence, the model learns a time-averaged transaction rate, which is expected to introduce a mild downward bias in future order volume predictions. Future work could address this by incorporating non-stationary extensions of the BG/NBD model or by applying time-decay weighting to calibration observations.

Third, both probabilistic models are structurally designed for repeat transactors and are not applicable to one-time or newly onboarded sellers who lack sufficient transaction history. These sellers, who represent 18.5% of the calibration sample, are excluded from the Gamma-Gamma estimation entirely. Complementary models, such as Pareto/NBD variants designed for sparse data, or machine learning approaches for cold-start sellers, could be integrated to extend population coverage.

Fourth, the proposed A/B testing protocol currently lacks an A/A testing phase prior to experimental assignment. Without A/A validation, pre-existing systematic differences between seller cohorts cannot be ruled out, which may compromise the causal interpretation of observed treatment effects. Future implementations should incorporate A/A testing as a prerequisite step. In addition, the A/B testing framework does not yet specify concrete numerical thresholds for key performance indicators such as minimum detectable effect sizes, target statistical power, or expected commission lift that would be required to operationalize the experimental design in practice.

Looking ahead, several directions offer meaningful extensions to the present framework. Incorporating seasonality and marketplace growth dynamics into the transaction model would reduce the stationarity bias and improve forecast calibration over longer horizons. Developing parallel modeling pipelines for one-time and new sellers would allow the SeLTV framework to cover the full seller population rather than the repeat-transactor subset. Finally, augmenting the framework with causal inference methods which can be supported by machine learning or deep learning approaches would enable the platform to move beyond descriptive segmentation toward identifying which seller-level and platform-level variables genuinely drive the key outcome metrics, providing a more rigorous evidential basis for strategic interventions.

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